

Exploring Beyond the Helmholtz Coils for Uniform Magnetic Field Generation with Topology Optimization

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Uniform magnetic fields are desired in a wide range of applications. The most known device achieving such a purpose is the pair of Helmholtz coils. It consists of two identical circular coils, with their main axis aligned and placed at a distance equal to their common radius from each other. The goal of this research is to find another arrangement of the coils in space so as to achieve the largest region possible with a uniform magnetic field. A topology optimization framework is developed for this purpose. It investigates the ideal distribution of current densities in an axisymmetric 2D domain. The designed topology provides a significant improvement of the volume with a uniform field, which is 6.37 times larger than the reference one obtained with Helmholtz coils.

Index Terms—computational design, actuator, magnetostatics, magnetic manipulation

I. INTRODUCTION

THE ability of generating a uniform magnetic field inside a Region Of Interest (ROI) is harvested in a wide range of applications such as magnetometers [1], magnetic resonance imaging [2], and micro-robot actuation [3], [4]. The most famous device achieving such a purpose is the pair of Helmholtz coils [5]. It consists of two identical circular coils, with their main axis aligned and placed at a distance equal to their common radius from each other. Despite being fairly simple, the span of the Region with a Uniform Magnetic Field (RUMF) is rather limited. In the literature, the size of this RUMF has been improved with additional coils placed in specific manners. Among them, the Maxwell tri-coils [6] configuration is the most known, but other arrangements were explored too [7]–[9]. Their main advantage is the improved field uniformity along the main axis. However, in practice, the desire for a uniform field is not restricted to the main axis but to an actual volume. The different existing configurations have poor uniformity performances when evaluated farther away from their main axis. Another aspect often overlooked when designing such a device, is the total electromagnetic potential (i.e. the number of turns times the electric current) that is not kept constant, resulting in systems with more components and a larger footprint.

The goal of this work is to find a novel coil topology which generates a larger RUMF, and not simply its span over the main axis. A topology optimization method is developed for this purpose. For a fair comparison with Helmholtz coils, the total electromagnetic potential is fixed. The design problem is restricted to the cross-section of the coils, assuming axisymmetry. This can be seen as the investigation of the spatial distribution of the circular turns constituting the coils.

This work is organized as follows: in Section II the geometry of the problem is defined with the physics and metrics used to evaluate the performance. In Section III the implemented topology optimization method is described. The results are then presented in Section IV, to end up with some discussions on the potential improvements and future work in Section V.

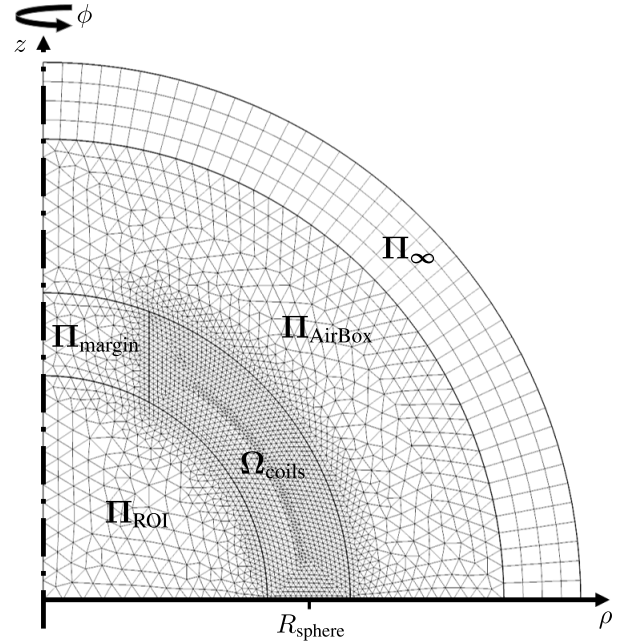


Fig. 1. Cross-section of the axisymmetric design problem. Only the upper half is shown because of planar symmetry. The mesh is composed of 5087 elements.

II. PROBLEM DEFINITION

A. Design Domain

The design problem is shown in Fig.1. The cylindrical coordinate system $\{\rho, \phi, z\}$ and the various sub-domains are shown as well as the mesh used for the computation. Only one-half of the 2D revolved domain is necessary because the problem is symmetric with respect to the (ρ, ϕ) plane. Π_{ROI} represents the ROI where a uniform field is desired. Ω_{coils} is the design domain where the optimal distribution of the coils is investigated. Π_{margin} is an empty domain to enable access to the ROI to insert samples or instruments from outside. Π_{AirBox} represents an *air-box* around the coils, and Π_{∞} a domain of infinity elements. The shape of the design domain Ω_{coils} is determined so as to include the spherical footprint of the

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reference Helmholtz coils (i.e. the smallest sphere into which the device is fully contained). Such choice is critical for 3D-systems as they are constituted of three sub-systems (one for each axis of the Cartesian space), individually scaled up so as to contain the previous one [5]. This nesting of the systems is permitted as long as the spherical footprint does not change. Otherwise, the total footprint of the 3D system is degraded. The mean radius of the spherical domain is taken as the radius from the center of the coordinate system, to the middle of the Helmholtz coil section. It can be calculated as:

$$R_{\text{sphere}} = \sqrt{R_H^2 + \left(\frac{R_H}{2}\right)^2}, \quad (1)$$

with R_H the radius of the circular Helmholtz coils. The width of the design space exactly contains the squared cross-section of the Helmholtz coils defined by the parameters presented below as shown in Fig.2b.

In order to perform a fair comparison, the target magnetic field inside the RUMF is fixed at $B_0 = 10$ mT, and the total electromagnetic potential to $\theta_{\text{tot}} = 1000$ Amps·turns. These requirements impose $R_H = 44.5$ mm for the radius of the Helmholtz coils. The wire diameter constituting the coils is chosen as $d_w = 1.15$ mm (17 AWG American Wire Gauge). The input current is fixed at $I = 4$ A, corresponding to a current density of $J_{\text{max}} = 3.8$ A/mm². The total number of turns $N = 250$ deduced from the fixed electromagnetic potential and input current, results in 125 turns in each coil constituting the pair of Helmholtz coils.

B. Physics and metrics

The magnetostatic problem is solved with finite element analysis. Ampere's law: $\nabla \times \mathbf{H} = \mathbf{J}_{\text{ext}}$ is applied on the full design domain to obtain the magnetic field $\mathbf{H}$ generated by an external current density $\mathbf{J}_{\text{ext}}$. The magnetic flux density is computed using $\mathbf{B} = \mu_0 \mathbf{H}$, assuming that everything is performed in air. The coils are not physically modeled but implicitly considered with the external current density distribution, so that no material definition or interface is needed. As the problem is axisymmetric, the physical values are defined as follows using the cylindrical coordinate system $\{\rho, \phi, z\}$: $\mathbf{J}_{\text{ext}} = (0, J^\phi, 0)$, resulting in $\mathbf{H} = (H^\rho, 0, H^z)$ and $\mathbf{B} = (B^\rho, 0, B^z)$. The planar symmetry is assured for a magnetic field being normal to the ρ -axis. This is imposed through the following boundary condition: $\mathbf{n} \times \mathbf{H} = 0$, with $\mathbf{n}$ the vector normal to the boundary.

Three metrics are used in this work to evaluate the performance of the different configurations. The two first ones are the relative spans with uniform field along the z -axis: δ_{uni}^z , and along the ρ -axis: δ_{uni}^ρ . They are defined as the ratio of the total length with a uniform field over the total span covered by the total mechanical footprint of the Helmholtz coils configuration for each axis individually:

$$\delta_{\text{uni}}^z = \frac{L_{\text{uni}}^z}{R_H} \quad ; \quad \delta_{\text{uni}}^\rho = \frac{L_{\text{uni}}^\rho}{2R_H} \quad (2)$$

The lengths with uniform field: L_{uni}^z , L_{uni}^ρ are defined as the lengths on each axis with a uniformity error inferior to 1%:

$$\left\{ L_{\text{uni}}^z : \left| \frac{B^z(\rho = 0, z)}{B^z(\rho = 0, z = 0)} - 1 \right| < 0.01 \right\}, \quad (3)$$

$$\left\{ L_{\text{uni}}^\rho : \left| \frac{B^z(\rho, z = 0)}{B^z(\rho = 0, z = 0)} - 1 \right| < 0.01 \right\}. \quad (4)$$

The last metric is on the relative volume with a uniform field δ_{uni}^V . It is defined as the ratio between the volume with uniform field over the spherical volume covered by the Helmholtz coils configuration:

$$\delta_{\text{uni}}^V = \frac{V_{\text{uni}}}{\frac{4\pi}{3} \cdot R_{\text{sphere}}^3}. \quad (5)$$

The volume inside the ROI with a uniform field: V_{uni} is defined as the volume with a uniformity error inferior to 1%:

$$\left\{ V_{\text{uni}} : \left| \frac{B^z(\rho, z)}{B^z(\rho = 0, z = 0)} - 1 \right| < 0.01 \right\}. \quad (6)$$

III. TOPOLOGY OPTIMIZATION

A standard density-based topology optimization method is implemented [10, ch. 1.1]. The control variables are artificial densities x defined onto the mesh of the design domain, which implicitly describe the topology:

$$x = x(\rho, z), \quad \text{with } \{\rho, z\} \in \Omega_{\text{coils}}. \quad (7)$$

The densities vary continuously from 0 to 1, corresponding to the absence or presence of material respectively. Intermediate values are referred to as *intermediate-materials*. This relaxation of the discrete design problem is required to rely on fast gradient-based optimization methods. It enables a realistic computational time considering the large number of design variables, and that a finite element analysis is performed every iteration. The control variables are not directly mapped to the current densities but require some preprocessing to prevent checkerboard patterns, impose a minimal feature size within the topology, and obtain a crisp topology without *intermediate-materials* [11]. First, a filter is applied by solving the following partial differential equation $\tilde{x} = x + R_{\text{filt}} \nabla^2 \tilde{x}$, with R_{filt} the filter radius (i.e. the minimal feature size), here set to the radius of the wire $d_w/2$ constituting the coils. The filtered variables are then projected to obtain a crisper definition of the topology using a hyperbolic tangent projection scheme [12]. It is defined by a projection slope β and a projection point at 0.5:

$$\hat{x} = \frac{\tanh(\beta(\tilde{x} - 0.5)) + \tanh(0.5\beta)}{2 \cdot \tanh(0.5\beta)}. \quad (8)$$

The last step is the mapping of the variables to the current densities. This is performed with a conventional Solid Isotropic Material Interpolation (SIMP) [10, ch. 1.1] scheme with a penalty factor q :

$$J^\phi = \hat{x}^q \cdot J_{\text{max}}. \quad (9)$$

The magnetic field can thus be solved with Ampere's law, directly leading to the magnetic flux density. The optimization problem can be formulated as follows:

$$\min_x : \iint_{\Pi_{\text{ROI}}} \left(\frac{B^z(\rho, z)}{B^z(\rho = 0, z = 0)} - 1 \right)^2 d\rho dz \quad (10a)$$

$$\text{s.t. : } 0.99 B_0 \leq B^z(\rho = 0, z = 0) \leq 1.01 B_0 \quad (10b)$$

$$0.99 \theta_{\text{tot}} \leq \iint_{\Omega_{\text{coils}}} J^\phi(\rho, z) d\rho dz \leq 1.01 \theta_{\text{tot}} \quad (10c)$$

$$x_{\text{avg}} \leq V_{\text{frac}} \quad \text{and} \quad 0 \leq x \leq 1. \quad (10d,e)$$

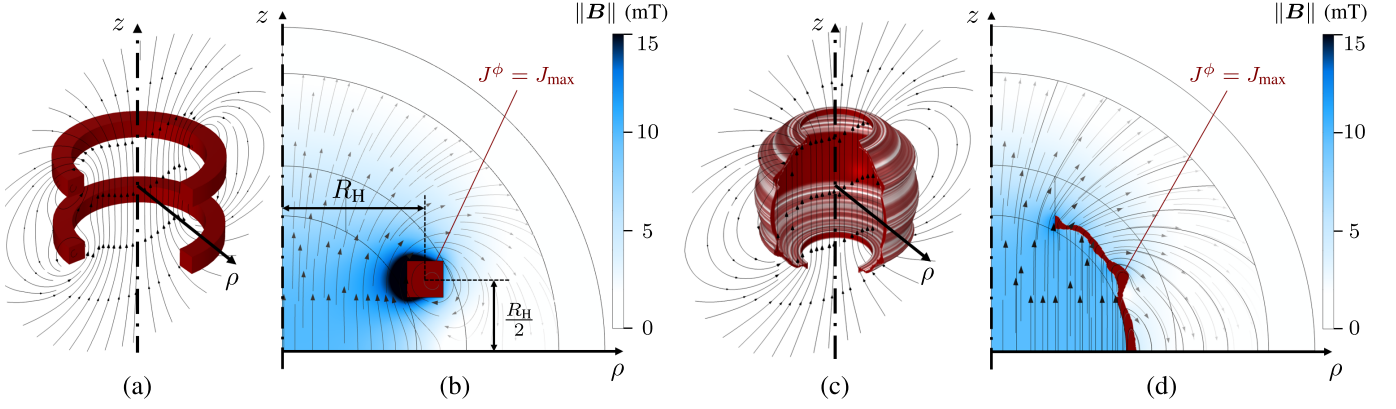


Fig. 2. Comparison of the reference and the proposed system: (a), (c) 3D cut view of the reference Helmholtz coils and the topology optimized configuration respectively; (b), (d) Magnetic flux density distribution in upper half of the 2D revolved domain for the Helmholtz coils and topology optimized configuration respectively.

The objective function (10a) is defined as the integration over the ROI of the squared error between the z -component of the magnetic flux density evaluated inside the domain, and the one evaluated in the middle point. The minimization of this function assures the maximization of the field uniformity inside the ROI. The magnitude of the magnetic flux density at this middle point is enforced at a target value B_0 with the explicit constraint (10b). As stated in Section II, the total electromagnetic potential θ_{tot} is kept constant with the second constraint (10c). A volume limitation V_{frac} of the average density x_{avg} , commonly used in topology optimization methods, is also added so as to have a final topology made of *full-materials* only ($J^\phi = J_{\text{max}}$ or $J^\phi = 0$). It is shown in (10d,e) next to the upper and lower bounds of the control variables x . The sensitivity analysis (i.e. computation of the gradients of the objective function and constraints with respect to the control variables) is performed with the adjoint variable method [10, ch. 1.2].

IV. RESULTS

The proposed method is implemented and solved using the commercial software COMSOL[®]. A conventional continuation scheme on the projection slope β and penalty factor q is implemented to prevent convergence to poor local minimum [10, ch. 1.1]. The specific requirement of a constant electromagnetic potential makes it more appealing for the optimizer to use *intermediate-materials* despite the penalization of (9). That is why a continuation scheme is also applied to the volume fraction constraint. The optimization is first performed for a relatively large value of V_{frac} . It is then continuously decreased to a smaller value so as to remove the remaining *intermediate-materials*, and obtain a topology with only extreme values $J^\phi = 0$ or $J^\phi = J_{\text{max}}$. Thus, multiple optimizations are repeated (the last solution of one optimization is the initial value for the next one) following the continuation scheme described in Table I. There are 2388 design variables, all initialized at $x=0.1$. The Method of Moving Asymptotes (MMA) [13] optimization solver is used with a maximum number of iterations set to 50 per optimization.

TABLE I
HYPERPARAMETERS CONTINUATION SCHEME, WITH β THE PROJECTION SLOPE, q THE PENALTY TERM, AND V_{frac} THE VOLUME FRACTION.

Optimization #	1	2	3	4	5	6	7	8	9
β	2	4	6	8	10	12	14	16	18
q	1	2	3	4	5	6	7	8	9
V_{frac}	0.3	0.3	0.3	0.3	0.3	0.25	0.20	0.18	0.16

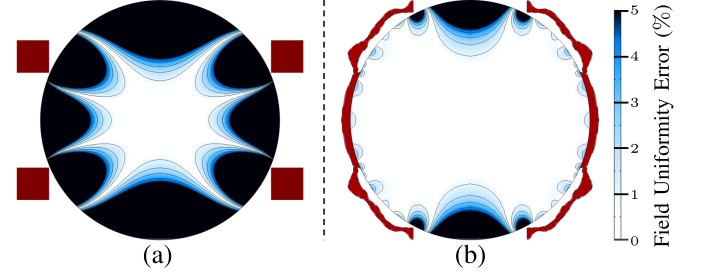


Fig. 3. Uniformity error inside the ROI: $|B^z(\rho, z) / B^z(\rho=0, z=0) - 1|$ with the cross-section of the coils in reddish brown for (a) the Helmholtz coils and (b) the topology optimized ones.

The full topology resulting from the complete optimization is shown in Fig. 2c. The corresponding distribution of current densities, and generated magnetic flux densities inside the 2D design domain are shown in Fig. 2d. Similarly, the Helmholtz coils configuration is shown in Fig. 2a and 2b. One can already see that the magnetic flux density looks much more uniform inside the ROI for the topology optimized coils. This is confirmed in Fig. 3, where the uniformity error is shown for the two configurations inside their ROI. The RUMF (with an error less than 1%) of the proposed topology optimized coils is much larger than the reference Helmholtz coils.

The z -component of the magnetic flux density along the different axes is shown in Fig. 4 for the Helmholtz and topology optimized configurations. The target field is attained, and the RUMF of the proposed topology is confirmed to be larger than the Helmholtz coils along the z and ρ axes.

The resulting topology satisfies the constraints with a value of magnetic flux density in the center point of

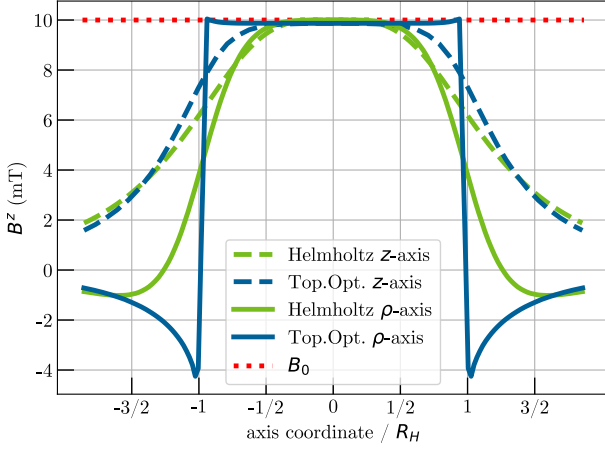


Fig. 4. Magnetic flux density distribution along the main z and radial ρ axes for the Helmholtz coils and the topology optimized (Top.Opt.) ones.

TABLE II
FIELD UNIFORMITY COMPARISON FOR DIFFERENT CONFIGURATIONS
WITH THE SAME MAGNETIC FLUX DENSITY $B_0 = 10$ mT, AND THE SAME
ELECTROMAGNETIC POTENTIAL $\theta_{\text{TOT}} = 1000$ AMPS-TURNS

	Helmholtz Coils	Maxwell tri-Coils	2D Top.Opt.
δ_{uni}^z	0.62	0.88	1.01
δ_{uni}^ρ	0.38	0.50	0.90
δ_{uni}^V	0.08	0.14	0.51

$B^z(\rho = 0, z = 0) = 9.9$ mT, and a total electromagnetic potential of $\theta_{\text{tot}} = 1009$ Amps-turns. The metrics evaluating quantitatively the performance of the different configurations are shown in Table II. The performance of the Maxwell tri-coils has also been added to provide more comparison insights. Its size has been scaled properly so as to match the requirements on the magnetic flux density target and the available electromagnetic potential. The proposed new topology performs better in all three metrics. The relative span with uniform field along the main axis: δ_{uni}^z , is 1.63 times larger than the Helmholtz coils, and 1.15 times larger than the Maxwell tri-coils. Furthermore, the relative span along the ρ axis: δ_{uni}^ρ , is 2.37 times larger than the Helmholtz, and 1.80 times larger than the Maxwell configurations. Finally, regarding the relative volume with a uniform field: δ_{uni}^V , the proposed topology achieves a volume 6.37 times larger than the Helmholtz coils, and 3.64 times larger than the Maxwell tri-coils. This confirms that the proposed method enables a vast improvement of the size of the RUMF.

V. DISCUSSIONS

Integrating the possibility of having negative current densities in the design domain (i.e. turns wound in the opposite direction) could improve the performance. However, as such current would contribute against the desired field direction, the required electromagnetic potential to achieve the target field will likely increase, degrading the footprint of the device.

As one can see, the resulting topology is fairly more complicated than the Helmholtz coils or the Maxwell tri-coils.

However, manufacturing such topology is feasible, as much more complex shapes have been fabricated in the past for the realization of gradient-coils [14]. The exact reconstruction of the turns distribution is beyond the scope of this work, but the filter radius assures that a wire can fit into any feature of the designed topology. Manufacturing errors could be accounted for by implementing a robust filtering process similar to [12].

Future work for this research is breaking the axisymmetry and investigating non-circular topology for a complete device more compact than a 3D-Helmholtz coils arrangement.

VI. CONCLUSION

This work proposes a new topology for uniform magnetic field generation. It is designed with a density-based topology optimization method, minimizing the squared error between the generated magnetic flux density, and the one evaluated at its central point. Two constraints enforce a fixed electromagnetic potential and a fixed magnitude of the target magnetic field. For the same spherical footprint, the proposed device achieves great performance. Its region with a uniform field has a volume more than 6.3 times larger than the reference Helmholtz coils.

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